

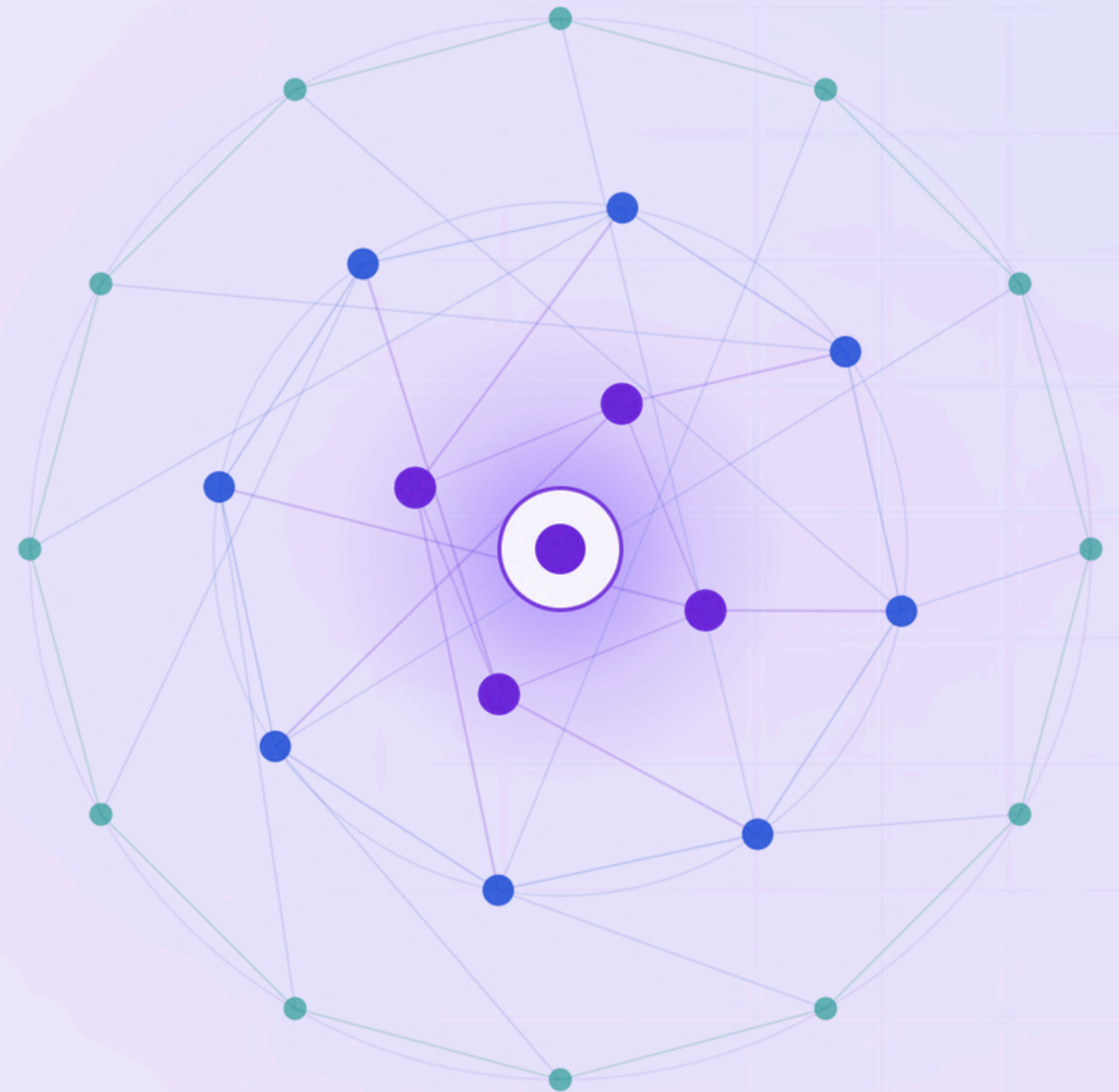
SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - **Theme 2 · Smart Guided Troubleshooting Engine**
- Team Name - **Thapar_Nexora_2**
- College Name - **Thapar Institute of Engineering and Technology , Patiala**
- Member Name & Email 1- **Chintan Sood , csood_be24@thapar.edu**
- Member Name & Email 2- **Avi Garg , aavi_be24@thapar.edu**
- Member Name & Email 3- **Shreshth Garg , sgarg7_be24@thapar.edu**
- Member Name & Email 4- **Khushi , Kkhushi1_be24@thapar.edu**
- Submission Github link -

https://github.com/khushichhabra7921/PRISM_GENAI_HACKATHON_Y2026_UPDATED_GITHUB



<300 ms on known issues



<8 s on new issues



0 invented steps



0 invalid deeplinks

Theme

Theme 2 · Troubleshooting: turn a TechCorp Nexa complaint and its SIIS article into a deeplinked, one-tap plan.

People describe **symptoms**.
Settings speak in **menus**.

Official kit: 20 complaints in `input.txt`, each arriving with the SIIS article `{title, content}` retrieved for it (11 distinct articles). The answer must match `sample_output.json`.

TOUCH · DISPLAY

“My Nexa X1 screen inputs are delayed and the touch responsiveness is laggy...”

input.txt · row_21 · SIIS: Touchscreen issues on a smartphone or tablet

SCREEN DAMAGE

“My Nexa Fold X1 screen is cracked again right where it folds.”

input.txt · row_19 · SIIS: Cracked or bleeding screen on a smartphone or tablet

DISPLAY SIZE

“My new smartphone’s main screen stays small and doesn’t fill the whole display...”

input.txt · row_8 · SIIS: Screen mirroring to your TechCorp TV

TODAY

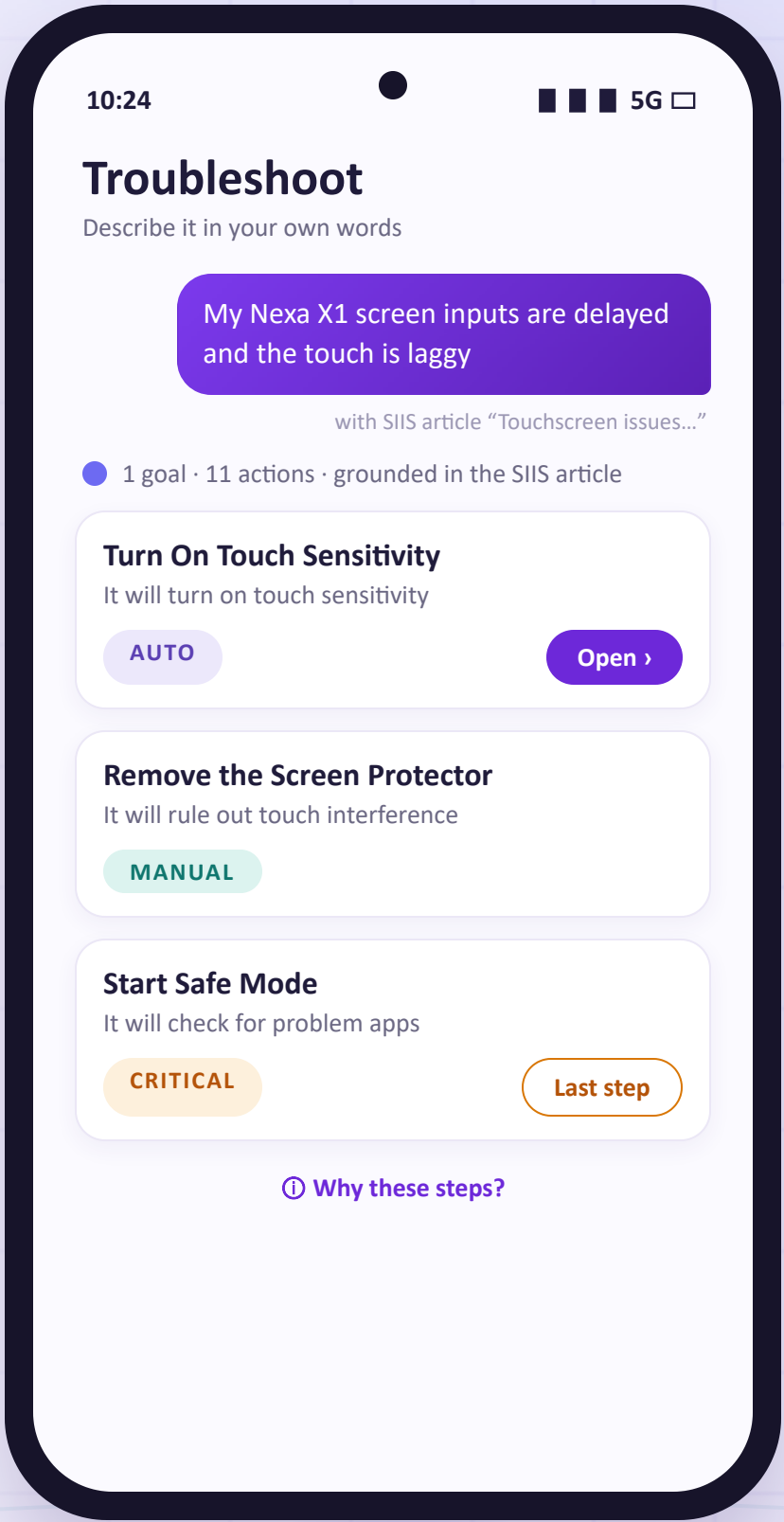
~15 min

manual triage per scenario: agent reads SIIS, orders steps, user hunts through Settings

OUR TARGET

<300 ms

known issues via the semantic cache; < 8 s for new ones, fully grounded in the article



Existing Solutions & Gaps

Each product owns part of the job. None turns a vague complaint into a grounded, ordered, validated plan with the exact Settings screen.

Solution	Good at	Natural language	Trouble-shooting	Ordered plan	Exact deeplink	Grounded + validated	Gap vs. our target workflow
Samsung Bixby	natural-language assistant; runs device actions	Core	Partial	—	Partial	—	executes actions; doesn't plan from grounded knowledge
Device Care	battery, storage, memory and device health	Partial	Partial	Partial	Partial	—	diagnoses state; no complaint → ordered plan
Samsung Members	diagnostics, support content, error reports	Partial	Core	Partial	Partial	—	user still has to find the right path
Warranty & Care	Bixby help, symptom search, self-diagnostics	Core	Core	Partial	Partial	—	no plan validation or deeplink layer
Samsung Help Center	deep library of issue-specific articles	Partial	Core	—	—	—	retrieval only; user turns text into taps
Google Pixel Help	NL help, device health, diagnostics	Core	Core	Partial	Partial	—	no Samsung SIIS grounding, deeplinks or cache
Microsoft Get Help	guided troubleshooters and diagnostics	Core	Core	Core	Partial	—	built for Windows, not Galaxy Settings
FAQ / web search	broad coverage, easy to find	Partial	Core	—	—	—	user interprets, orders and executes
Human / remote support	expert diagnosis for unusual cases	Core	Core	Core	—	—	slow and costly; not instant or deterministic
Our engine	complaint → verified one-tap plan	✓ Core	✓ Core	✓ Core	✓ Core	✓ Core	the full chain, with an evidence trail



Assistants act...
but don't plan from trusted knowledge



Diagnostics measure...
but don't explain what to do next

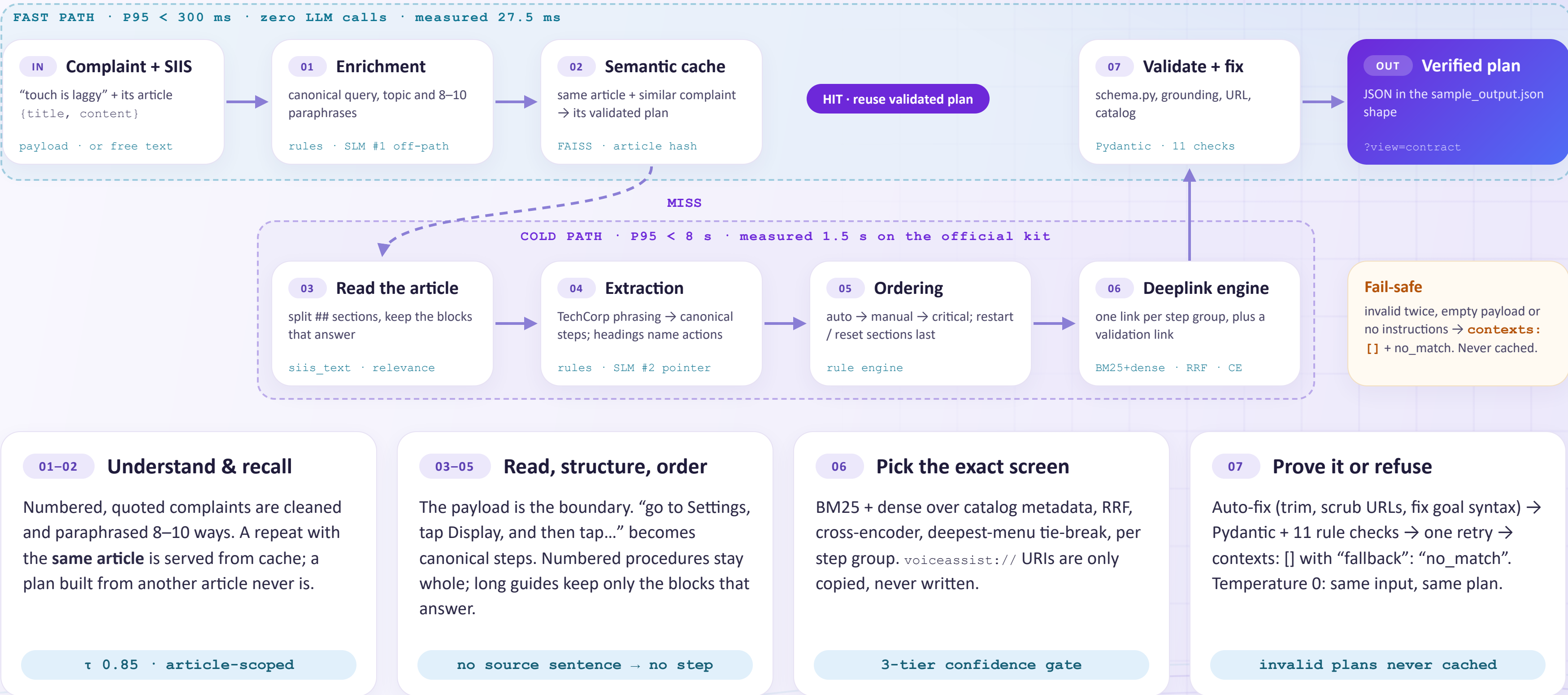


Articles explain...
but leave every tap to the user

We connect intent → knowledge → action.
Bixby stays the execution layer.

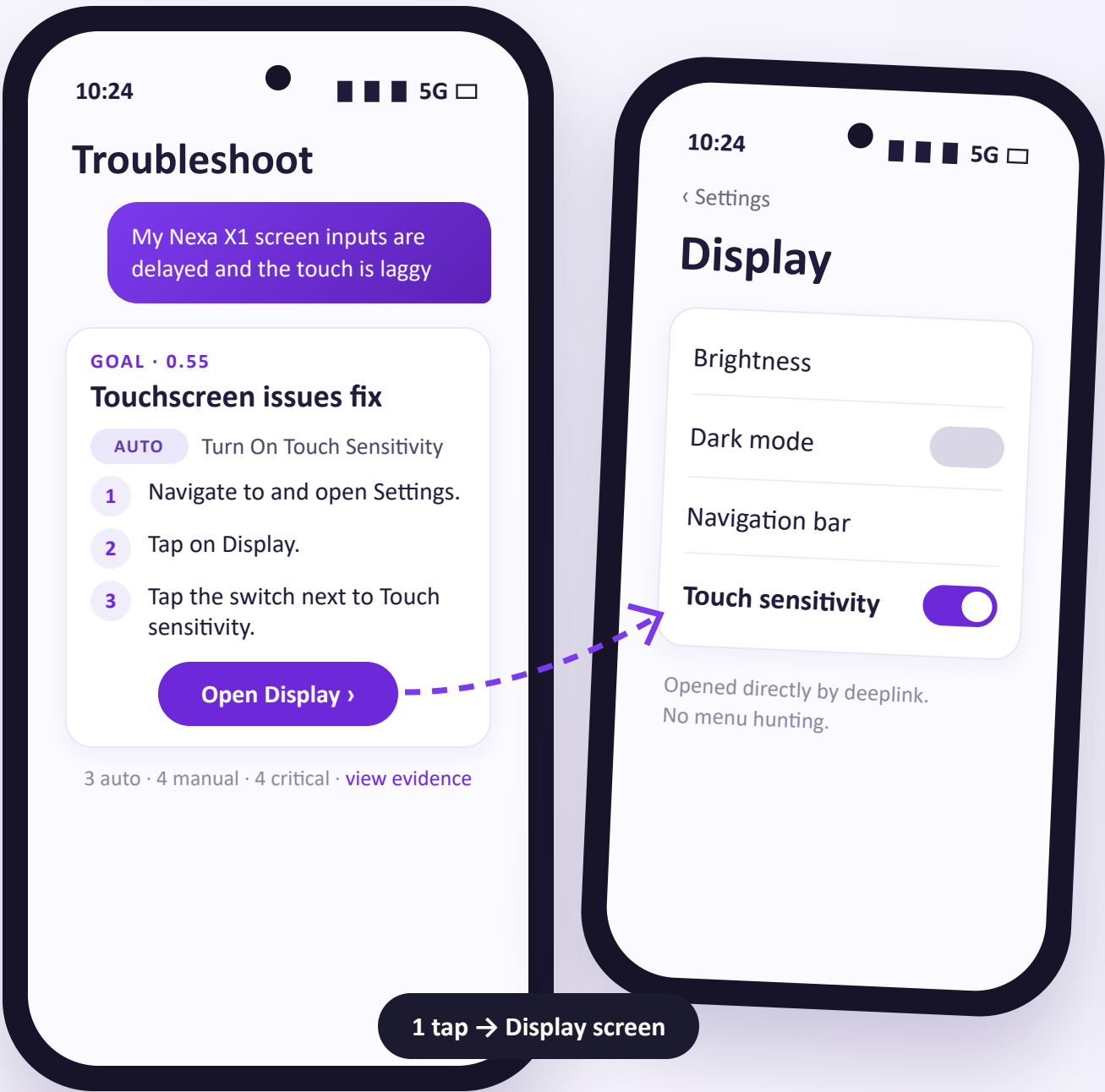
Our Solution & Architecture Diagram

LLMs reason, retrieval grounds, deterministic code validates. The SIIS article sent with the complaint is the only source of steps.



Demo & Product Walkthrough

A real complaint from the official input.txt, sent with its SIIS article: what the user sees, what the API returns, and the receipt behind every action.



Demo video: <https://youtu.be/UWmbvZDs3Ao?si=QjdoR6tNjP7mBOxH>

```
sgte · /v1/troubleshoot 200 · 1.4 s cold

# request · data/input.txt row_21 + its SIIS payload
POST /v1/troubleshoot?view=contract
{ "query": "My Nexa X1 screen inputs are
    "delayed and the touch ...",
  "siis_response": { "title": "Touchscreen issues...",
    "content": "## 1. Factors..." } }

# response · 200 OK · shape of sample_output.json
{ "query": ..., "response": { "contexts": [{
  "goal": "Follow these steps to perform"
    "this Touchscreen Issues Troubleshooting",
  "title": "Touchscreen issues fix", "score": 0.55,
  "actions": [{
    "actionName": "Turn On Touch Sensitivity",
    "description": "It will turn on touch sensitivity",
    "stepGroups": [{ "steps": [ ...,
      "Tap on Display.", ... ],
    "actionableDeeplink": {
      "deeplink": "voiceassist://masked/act/b8a0562dd6",
      "description": "Open display settings",
      "originalType": "onURL" },
    "validationDeeplink": null }],
    "category": "auto" }, ... 10 more ]}}]}
```

- 1 **Complaint + article in**
“My Nexa X1 screen inputs are delayed...” + SIIS
“Touchscreen issues on a smartphone or tablet”
- 2 **Read**
numbered procedure kept whole: 10 of 10 blocks, 42
steps copied from the text
- 3 **Planned**
11 actions: 3 auto → 4 manual → 4 critical; least
disruptive first
- 4 **One tap**
each Settings navigation carries its own catalog deeplink
- 5 **Recalled**
same complaint + same article again: 28 ms (P95), \$0, no
LLM call

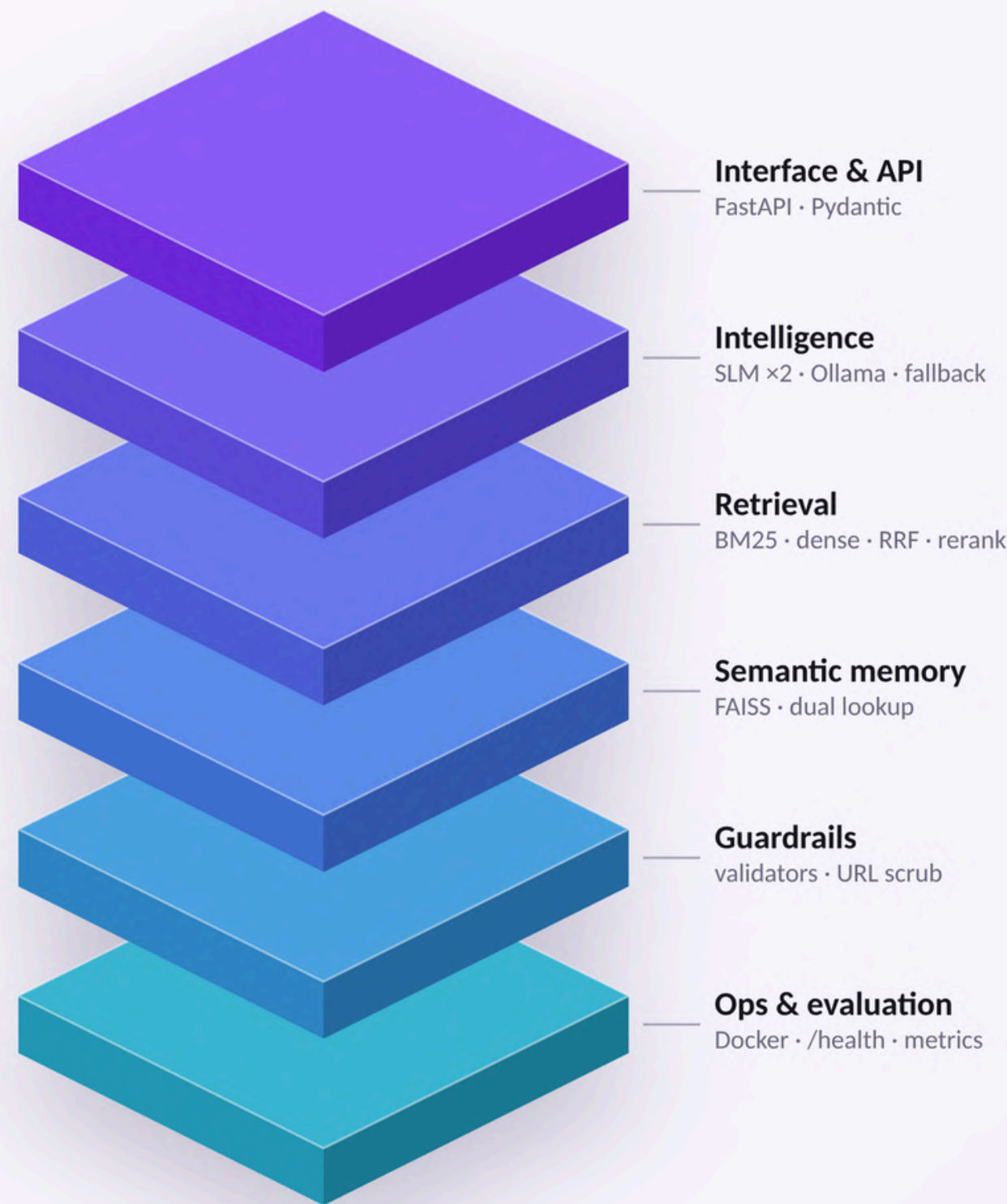
EVIDENCE TRAIL · PER ACTION

- SIIS section
- block relevance
- bm25 · dense
- runner-up

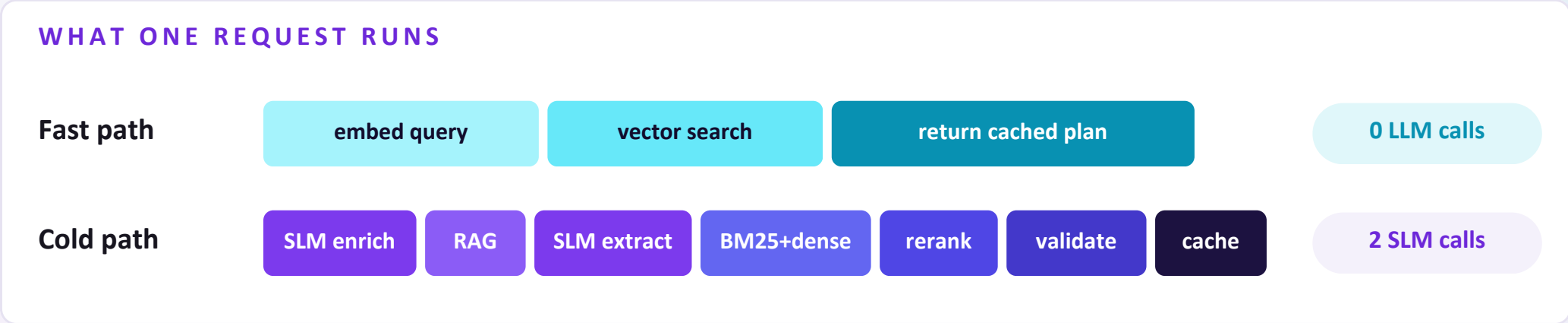
- source sentence
- kept/dropped
- rerank · rrf
- confidence tier

Tools and Tech Stack Used

A layered, containerised, SLM-first stack. Every layer is there for a latency or safety reason.



Layer	What we run	Why this, not the obvious alternative
Interface	FastAPI · Pydantic · schema.py	contract enforced at the boundary; JSON only
Intelligence	local SLM (Ollama) + hosted fallback · llm_client.py · temp 0	two narrow jobs don't need a frontier model per call
Retrieval	SIIS RAG · BM25 · dense · RRF · cross-encoder	keywords hit parent menus; embeddings miss exact terms
Memory	FAISS · dual lookup (raw + canonical) · pre-warmed	exact-string keys miss paraphrases
Guardrails	Pydantic validators · regex URL scrub · fuzzy grounding	prompts can't be trusted with word counts
Ops & eval	Docker Compose · /health · benchmark · metrics.md	numbers logged per request, not estimated



Impact & Use Case

Every Galaxy shipped is a future support conversation. Here is who benefits, and how the engine behaves in three real situations.

241.2M

Samsung smartphones shipped in 2025

~19% global share

~15 min

manual triage per scenario today

problem brief

61%

prefer self-service for simple issues

Salesforce, 2025

92%

would use a self-serve knowledge base

Higher Logic, 2024

HOW WE SIZE IT

time saved ≈ automation rate × 15 min × query volume

rate = measured cache-hit rate

A

Known issue, new wording

FAST PATH

“My Wi-Fi keeps dropping”

paraphrase hits the dual cache

validated plan, zero LLM calls

one tap to the exact Wi-Fi screen

B

Two problems, one sentence

MULTI-SYMPTOM

“Screen flickers and my battery dies”

split into Display + Battery

each runs the full grounded pipeline

two verified goals in one response

C

An issue SIIS doesn’t cover

SAFE FAILURE

complaint with no source text

contexts: [] + no_match, nothing invented

clustered offline with similar misses

gap report for the SIIS content team

📱

Users

one tap instead of menu hunting; slang and typos understood; both halves of a complaint answered

best for older & first-time users

🎧

Support teams

escalations arrive with the evidence trail; no-match clusters show where SIIS is thin

5,255 courses · 39,703 trainees in 2025

📝

Environment

fewer service visits for config-only issues; a cache hit makes zero LLM calls

SLM-first, proven by ablation

📈

Samsung

demonstrable AI safety; pattern ports to SmartThings & Galaxy Watch

3,000+ touchpoints · 12,500+ engineers (India)

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Innovation Highlights, Results and Limitations

Four capabilities that make the official kit work end to end, the numbers we measured on it, and where it can still fail.



01

Evidence trail

Every action carries its SIIS section, source sentence, block relevance and why the runner-up screen lost.

grounding you can show



02

Reads real support articles

Markdown sections, comma-chain menus and one-step-per-line lists become canonical steps and step groups.

built for the official kit



03

Article-scoped cache

A repeat with the same article returns in 28 ms; a plan from another article is never served.

fast and safe



04

Graded confidence

Three tiers instead of match-or-nothing: auto-attach, attach with a visible score, or fall back.

more coverage, same precision

RESULTS · OFFICIAL KIT, MEASURED

100%

schema-valid

20 official complaints

100%

pass 11 checks

0 violations

100%

steps grounded

308 steps in the payloads

100%

catalog deeplinks

9 links · 0 invented

P95 28 ms repeat

P95 1.5 s cold

0 URL leaks · \$0

Synthetic regression kit: gold step accuracy 3 / 3 · deeplink relevance 2 / 2 · 80.2% of unseen paraphrases served with 0 LLM calls · 69 tests pass.

LIMITATIONS → MITIGATIONS

No deeplink catalog in the official kit

→ synthetic catalog in the official scheme; the one real entry copied

No gold plans for the 20 complaints

→ contract + grounding measured; accuracy on synthetic gold

Cross-encoder poorly calibrated on support text

→ numbered procedures are kept whole

Article only partly about the complaint

→ keep the answering blocks; no_match if none

Whole procedures can be long (up to 11 actions)

→ ordered auto → manual → critical

1.5B SLM too slow on a laptop CPU

→ runs off the request path; hosted fallback

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What's Next

Where the architecture goes from here: broader coverage, a system that learns from use, and reach beyond the phone.

- **Broader device and settings coverage:** extend the deeplink index across more Galaxy models, One UI versions and settings paths. Every new deeplink passes the 11 validators before it is served, so users are never sent to a screen that does not exist on their device.
- **Confirmed-fix learning loop:** when a user confirms a step solved their problem, that signal is logged and fed back to retrain ranking and tune the match threshold τ . Proven fixes rise to the top, and τ is set from real outcomes rather than guesswork.
- **No-match clusters feed SIIS content:** queries with no confident match are grouped by intent, and the largest clusters are ranked by volume and handed to content authors as new guides. Every miss becomes a concrete content gap.
- **More query languages and registers:** add further languages and cover formal, casual and misspelled phrasing, keeping ranking quality consistent so users get the same accurate guidance however they describe the problem.
- **Hardened offline / on-device SLM:** run the core troubleshooting flow on the device without a connection, with a graceful fallback to the cloud path when a query needs it. Guidance stays available with weak or no network.
- **Cache that improves with use:** the semantic cache is pre-warmed from real query patterns, so frequent questions are served from cache and repeat queries stay under 300 ms even as traffic grows.

Brownie Points (Differentiation)

Every differentiator is backed by published evidence, not internal claims. Our own measured numbers go in metrics.md.

61%

of users prefer self-service over an agent for simple issues *Salesforce, 2025*

92%

would use a self-serve knowledge base if one were offered *Higher Logic, 2024*

70%

of enterprise AI tools expected to use RAG by end of 2025 *Gartner*



Sub-300 ms hits, sub-8 s cold

known issues never wait on an LLM

caching: up to 90% lower LLM cost, up to 250x lower latency (Redis / Percona, 2025)



Two-tier paraphrase cache

raw words, then the canonical query

61–69% of repeat queries served without a fresh LLM call (GPTCache, arXiv:2411.05276)



Hybrid retrieval, exact screens

BM25 + dense + RRF + cross-encoder

hybrid beats either method by 26–31% NDCG (BEIR aggregate)



Guardrails live in code

syntax, length, deeplinks, URL leaks

retrieval + validation: hallucinated steps ~21% → < 7.5% (NAACL 2024)



Fails safe, never invents

no source → contexts: [] + no_match

grounded generation: 70–90% less hallucination (RAG studies, 2025–26)



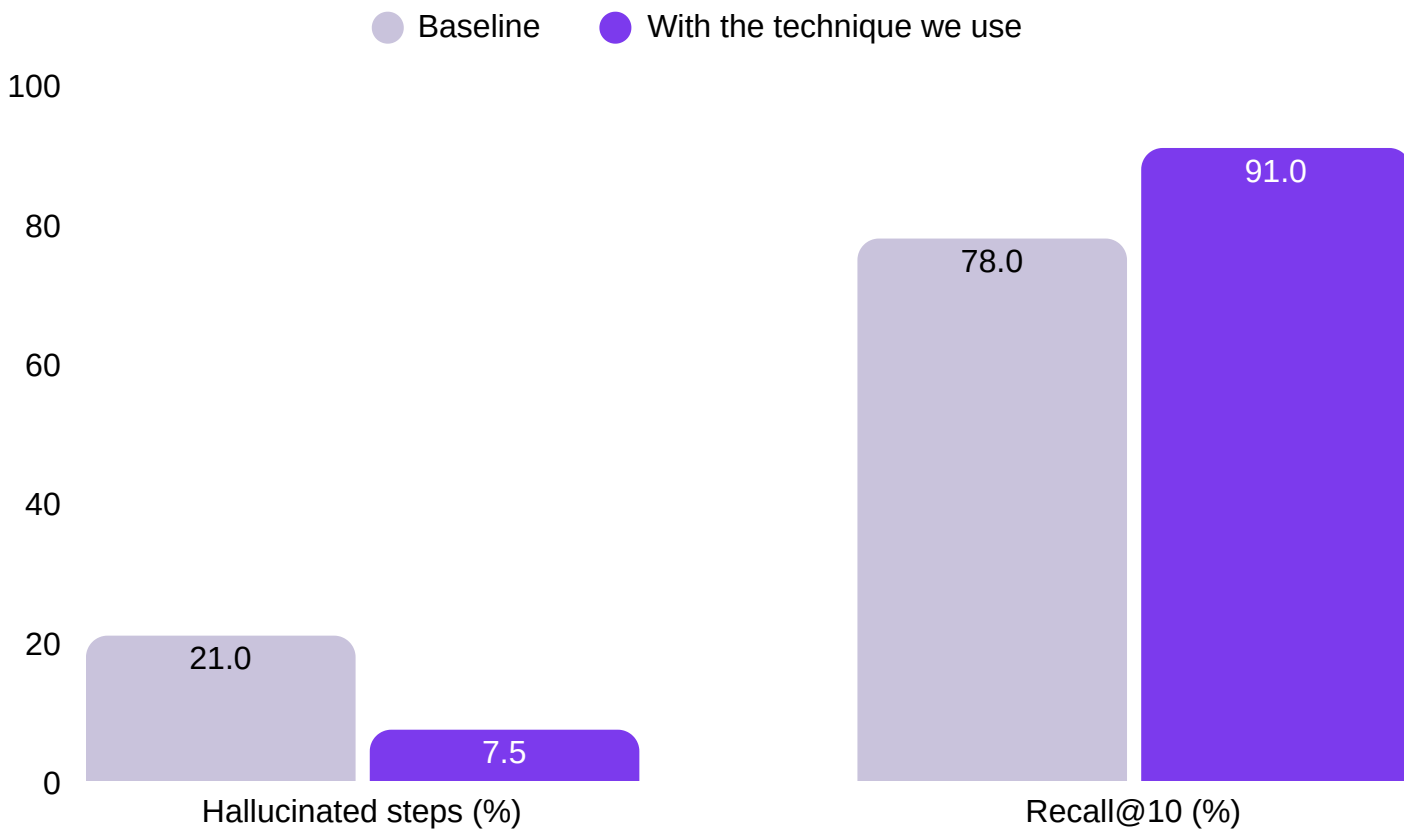
Tuned and measured

ablations, τ sweeps, false-hit rates

untuned fusion can trail a tuned baseline by ~10 MRR points (AIMultiple, 2025)

TWO RECEIPTS, CHARTED

Published results behind our design choices



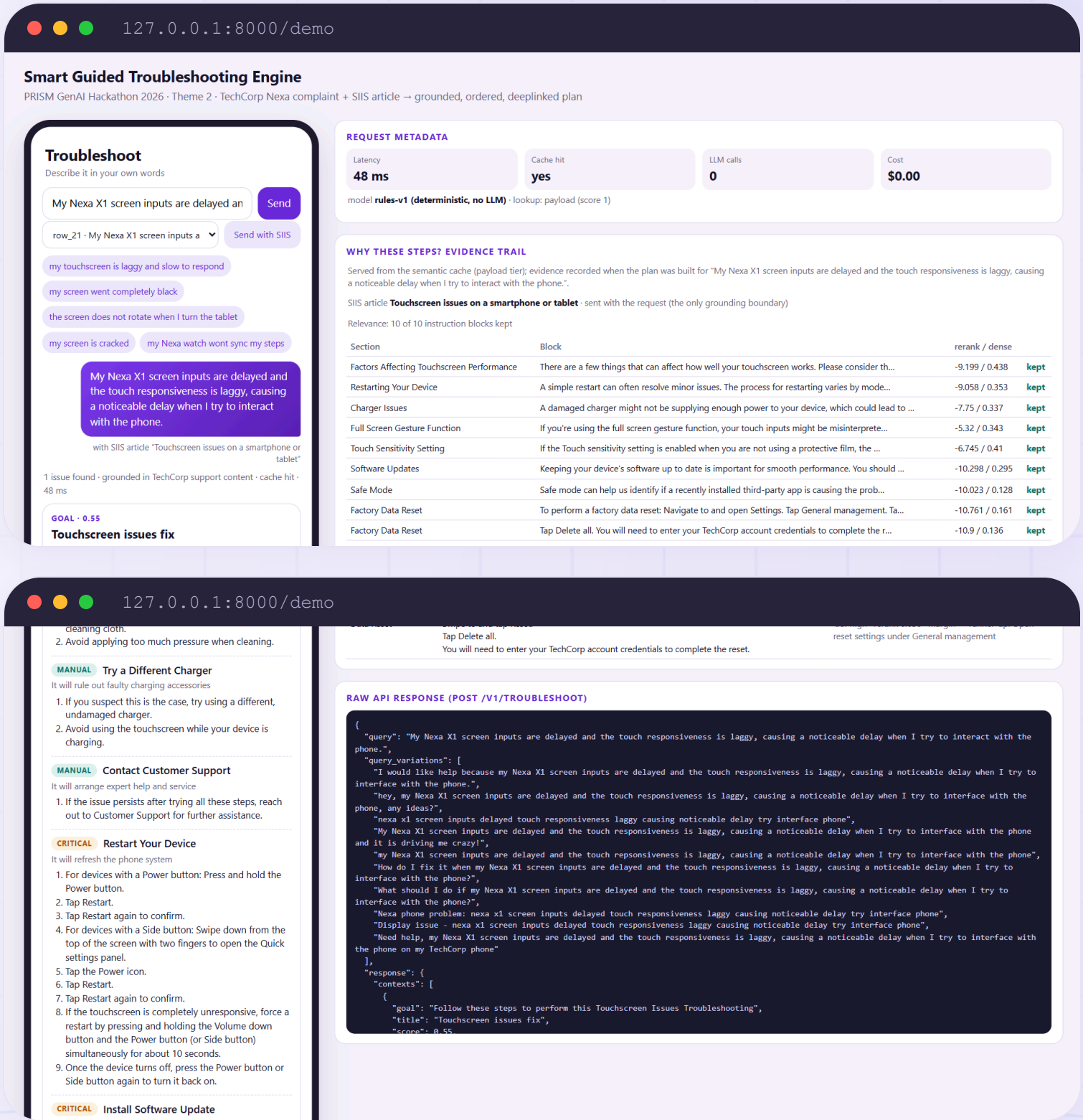
Left: ungrounded vs. retrieval + validation (Ayala & Bechard, NAACL 2024). Right: single retriever vs. hybrid (BEIR aggregate).

Checklist: Updated on Public GitHub

What judges open first. Every item below is live in the repository.

Deliverable	Status	Link / note
Working prototype code public or shared GitHub repo	Yes	<code>github.com/khushichhabra7921/PRISM_GENAI_HACKATHON_Y2026</code>
README reproducible setup instructions	Yes	<code>docker compose up</code>
Official-kit outputs 20 responses, sample_output.json shape	Yes	<code>outputs/ · bench/official.py</code>
Demo video max 5 minutes, YouTube or Drive	Yes	<code>https://youtu.be/UWmbvZDs3Ao?si=Qjd oR6tNjP7mBOxH</code>
Presentation PPT or PDF	Yes	<code>this deck</code>

LIVE DEMO · OFFICIAL KIT



Research and References

- **Luo, J., Li, T., Wu, D., Jenkin, M., Liu, S., & Dudek, G. (2024).** Hallucination Detection and Hallucination Mitigation: An Investigation. Samsung AI Center, Montreal. [arXiv:2401.08358](<https://arxiv.org/abs/2401.08358>)
- **RAG / Grounded Generation** Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020. [arXiv:2005.11401](<https://arxiv.org/abs/2005.11401>)
- **Shuster, K., Poff, S., Chen, M., Kiela, D., & Weston, J. (2021).** Retrieval Augmentation Reduces Hallucination in Conversation. [arXiv:2104.07567](<https://arxiv.org/abs/2104.07567>)
- **Nogueira, R. & Cho, K. (2019).** Passage Re-ranking with BERT. [arXiv:1901.04085](<https://arxiv.org/abs/1901.04085>) — the standard citation for cross-encoder reranking, justifying your rerank stage after RRF fusion.
- **Anonymous (2025). Winning Big with Small Models:** Knowledge Distillation vs. Self-Training for Reducing Hallucination in Product QA Agents. [arXiv:2502.19545](<https://arxiv.org/abs/2502.19545>)
- **Applied Domain — Troubleshooting via RAG** Khanda, R. (2024). Agentic AI-Driven Technical Troubleshooting for Enterprise Systems: A Novel Weighted Retrieval-Augmented Generation Paradigm. [arXiv:2412.12006]. (<https://arxiv.org/abs/2412.12006>)

Mentor : Shruti Aggarwal

Thank you

Understand

Ground

Plan

Navigate

Prove

Smart Guided Troubleshooting Engine Nexora · Thapar Institute of Engineering and Technology , Patiala

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India